Mail Server & Delivery System Integration in Kubernetes

Complete Implementation Report

Project: ShopFast E-Platform
Date: August 2026
Environment: Kubernetes (k3s) on Single-Node Cluster
Domain: shopfast.live

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1. Executive Summary

This document describes the complete implementation of a mail server and delivery system integrated into the ShopFast e-commerce platform running on Kubernetes. The system provides:

- **Inbound email reception** via Postfix SMTP (port 25)
- **Outbound email sending** via Postfix submission (ports 587/465)
- **Email storage** in mbox format at `/var/mail/`

- **Email access** via Dovecot IMAP (port 143)
- **Webmail interface** via RainLoop served at `https://shopfast.live/mail/`
- **DKIM signing** via OpenDKIM for email authentication

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2. Architecture Overview

2.1 High-Level Architecture

```
Internet → [Port 25/587/465] → Postfix (SMTP Server)
                                    ↓
                            Dovecot (Auth + IMAP)
                                    ↓
                            /var/mail/ (Storage)
                                    ↓
                            RainLoop (Webmail UI)
                                    ↓
                            User Browser (HTTPS)
```

2.2 Component Interaction

Component	Protocol	Port	Purpose
Postfix	SMTP	25	Inbound email reception
Postfix	Submission	587	Outbound email (STARTTLS)
Postfix	SMTPS	465	Outbound email (Implicit TLS)
Dovecot	IMAP	143	Email access for clients
Dovecot	Auth	2525	SASL authentication for Postfix
RainLoop	HTTP	80	Webmail interface
OpenDKIM	milter	8891	DKIM signature signing

2.3 Pod Architecture

The mail system runs in two pods:

mail-server pod (hostNetwork: true): - `postfix` container: Handles SMTP, queue management, delivery - `dovecot` container: Handles IMAP, SASL authentication

mail pod (RainLoop UI): - `mail` container: Nginx + PHP-FPM serving RainLoop webmail

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3. Component Details

3.1 Postfix Configuration

Image: `localhost:5000/shopfast/postfix:v25`

Key Configuration (`main.cf`):

```
myhostname = mail.shopfast.live
mydomain = shopfast.live
myorigin = $mydomain
mydestination = localhost, shopfast.live
relay_domains = shopfast.live

# SASL Authentication via Dovecot
smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = inet:localhost:2525

# Restrictions
smtpd_recipient_restrictions = permit_mynetworks, permit_sasl_authenticated, reject_u
smtpd_client_restrictions = permit

# Delivery Configuration
mailbox_command = /usr/local/bin/mbox-deliver "$USER"
queue_run_delay = 60s
mailbox_size_limit = 0
message_size_limit = 52428800

# TLS
smtpd_tls_cert_file = /opt/postfix-tls/cert.pem
smtpd_tls_key_file = /opt/postfix-tls/key.pem
smtpd_tls_security_level = may
```

Delivery Script (`/usr/local/bin/mbox-deliver`):

```
#!/bin/sh
FROM="From $SENDER $(date)"
{ echo "$FROM"; cat; } >> /var/mail/$1
```

This script appends incoming emails to the recipient's mbox file with the standard mbox `From` separator line.

3.2 Dovecot Configuration

Image: `localhost:5000/shopfast/dovecot:v1`

Key Configuration (`dovecot.conf`):

```
protocols = imap
listen = *, ::
auth_mechanisms = plain login
auth_username_format = %Ln

passdb {
  driver = pam
}

userdb {
  driver = passwd
}

mail_location = mbox:/var/mail/:INBOX=/var/mail/%u
mail_uid = 5000
mail_gid = 5000

service auth {
  inet_listener {
    port = 2525
  }
}

ssl = no
```

3.3 RainLoop Webmail

Image: `localhost:5000/shopfast/mail:v13`

Key Configuration:

- **basePath:** `/mail/` - Makes RainLoop emit `/mail/*` URLs
- **Domain config:** `shopfast.live.ini` with IMAP/SMTP settings
- **SSL verification:** Disabled (for self-signed certs)

Domain Configuration (`shopfast.live.ini`):

```
imap_host = mail-imap
imap_port = 143
imap_secure = 0
smtp_host = mail-smtp
smtp_port = 587
```

```
smtp_secure = TLS
smtp_auth = 1
```

3.4 OpenDKIM Configuration

Key Configuration (`opendkim.conf`):

```
Socket                  inet:8891@127.0.0.1
UserID                  opendkim
SigningTable            refile:/etc/opendkim/SigningTable
KeyTable                refile:/etc/opendkim/KeyTable
SignatureAlgorithm       rsa-sha256
```

DKIM DNS Record: - Selector: `shopfast2026` - Record: `shopfast2026._domainkey.shopfast.live` (TXT)

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4. Kubernetes Deployment

4.1 Namespace and Resources

All mail resources are deployed in the `shopfast` namespace:

```
apiVersion: v1
kind: Namespace
metadata:
  name: shopfast
```

4.2 Persistent Storage

PersistentVolume (`mail-storage.yaml`):

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: mail-storage
spec:
  capacity:
    storage: 5Gi
  accessModes:
    - ReadWriteOnce
  persistentVolumeReclaimPolicy: Retain
  storageClassName: manual
  hostPath:
```

```
    path: /var/mail
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: mail-storage
spec:
  storageClassName: manual
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 5Gi
```

4.3 Mail Server Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mail-server
  namespace: shopfast
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mail-server
  template:
    metadata:
      labels:
        app: mail-server
    spec:
      hostNetwork: true
      dnsPolicy: ClusterFirstWithHostNet
      containers:
        - name: postfix
          image: localhost:5000/shopfast/postfix:v25
          ports:
            - containerPort: 25
            - containerPort: 587
            - containerPort: 465
          volumeMounts:
            - name: mail-storage
              mountPath: /var/mail
            - name: postfix-tls
              mountPath: /opt/postfix-tls
          command: ["sh", "-c", "adduser -D -u 1001 -g users admin 2>/dev/null; exec /usr/libexec
        - name: dovecot
          image: localhost:5000/shopfast/dovecot:v1
          ports:
            - containerPort: 143
            - containerPort: 2525
          volumeMounts:
```

```
            - name: mail-storage
              mountPath: /var/mail
```

Key Design Decisions:

1. **hostNetwork: true** - Required for SMTP port 25 binding to node IP
2. **PersistentVolume** at `/var/mail` - Survives pod restarts
3. **TLS via Kubernetes Secret** - `mail-tls` secret with cert/key
4. **Admin user creation** at container startup - UID 1001 for mail delivery

4.4 Services

```
# IMAP and Auth services
apiVersion: v1
kind: Service
metadata:
  name: mail-imap
spec:
  selector:
    app: mail-server
  ports:
    - name: imap
      port: 143
      targetPort: 143
    - name: auth-tcp
      port: 2525
      targetPort: 2525
  type: ClusterIP

# SMTP services
apiVersion: v1
kind: Service
metadata:
  name: mail-smtp
spec:
  selector:
    app: mail-server
  ports:
    - name: smtp
      port: 25
      targetPort: 25
    - name: submission
      port: 587
      targetPort: 587
    - name: submissions
      port: 465
      targetPort: 465
  type: ClusterIP
```

4.5 Ingress Configuration

```
- path: /mail(/|$)(.*)
  pathType: Prefix
  backend:
    service:
      name: mail
      port:
        number: 80
```

The ingress rewrites `/mail/*` requests to `/` at the RainLoop pod, while RainLoop's basePath configuration generates `/mail/*` URLs back to the browser.

4.6 RainLoop Init Container

```
initContainers:
  - name: init-rainloop-settings
    image: busybox:1.36
    command:
      - sh
      - -c
      - |
        mkdir -p /data/_data_/_default_/storage/cfg/ad/admin@shopfast.live
        cp /settings/settings_local /data/_data_/_default_/storage/cfg/ad/admin@shopfast.live/set
        mkdir -p /data/_data_/_default_/configs
        cp /settings/application.ini /data/_data_/_default_/configs/application.ini
        mkdir -p /data/_data_/_default_/domains
        cp /settings/shopfast.live.ini /data/_data_/_default_/domains/shopfast.live.ini
```

---

5. Email Delivery Flow

5.1 Inbound Email (External → Admin)

```
1. External sender → SMTP connection to shopfast.live:25
2. Postfix accepts email, verifies recipient
3. Postfix queues email in /var/spool/postfix/
4. qmgr processes queue (every 60s)
5. local transport invokes mailbox_command
6. mbox-deliver script appends to /var/mail/admin
7. Dovecot serves email via IMAP to RainLoop
```

5.2 Outbound Email (Admin → External)

```
1. User composes email in RainLoop
2. RainLoop connects to mail-smtp:587 (STARTTLS)
3. Postfix verifies SASL auth via Dovecot (port 2525)
4. Postfix signs email with DKIM (OpenDKIM milter)
5. Postfix delivers to recipient's MX server
```

5.3 Queue Processing

Setting	Value	Description
queue*run*delay	60s	How often qmgr processes queue
minimal*backoff*time	60s	Minimum delay before retry
maximal*queue*lifetime	1d	Maximum time in queue

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6. DNS Configuration

6.1 Required Records

Type	Name	Value	Purpose
A	mail.shopfast.live	72.62.250.5	Mail server IP
MX	shopfast.live	mail.shopfast.live	Mail exchange
TXT	shopfast2026._domainkey.shopfast.live	(DKIM public key)	DKIM authentication
TXT	shopfast.live	v=spf1 mx ~all	SPF record

6.2 MX Record Configuration

```
shopfast.live.    IN  MX  10  mail.shopfast.live.
mail.shopfast.live.  IN  A  72.62.250.5
```

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7. Security Implementation

7.1 TLS Configuration

- **Certificate:** Managed by cert-manager with Let's Encrypt
- **Secret:** `mail-tls` in `shopfast` namespace
- **Mount:** `/opt/postfix-tls/` in Postfix container
- **Levels:**
  - Port 25: `may` (opportunistic TLS)
  - Port 587: `encrypt` (required STARTTLS)
  - Port 465: `wrappermode` (implicit TLS)

7.2 Authentication

- **SASL Type:** Dovecot
- **Mechanisms:** PLAIN, LOGIN
- **Port:** 2525 (TCP loopback)
- **Path:** `inet:localhost:2525`

7.3 Firewall Rules (UFW)

```
25/tcp   ALLOW  Anywhere    # SMTP
143/tcp  ALLOW  10.42.1.0/24  # IMAP (internal)
587/tcp  ALLOW  Anywhere    # Submission
465/tcp  ALLOW  Anywhere    # SMTPS
```

7.4 DKIM Signing

- **Selector:** shopfast2026
- **Algorithm:** rsa-sha256
- **Socket:** inet:8891@127.0.0.1
- **milter_protocol:** 6

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8. Troubleshooting Guide

8.1 Common Issues and Solutions

Issue	Cause	Solution
ConfigMap subPath caching	Kubernetes doesn't update subPath mounts	Bake config into Docker image
"mbox-deliver: not found"	Script missing from image	Add script to Dockerfile
Queue not processing	qmgr not triggering transport	Set queue*run*delay = 60s
Email stuck in queue	Permission issues	Ensure admin user exists (UID 1001)
"Domain is not allowed"	RainLoop config missing	Add domain config and basePath
Auth failure	SASL misconfiguration	Verify Dovecot auth on port 2525

8.2 Diagnostic Commands

```
# Check Postfix queue
kubectl exec -n shopfast <pod> -c postfix -- postqueue -p

# Check Postfix config
kubectl exec -n shopfast <pod> -c postfix -- postconf -n

# Test delivery manually
kubectl exec -n shopfast <pod> -c postfix -- postfix reload

# Check mail logs
kubectl exec -n shopfast <pod> -c postfix -- cat /var/log/maillog

# Verify IMAP access
kubectl exec -n shopfast <pod> -c dovecot -- doveadm auth test admin@shopfast.live Admin@123

# Check DKIM signing
kubectl exec -n shopfast <pod> -c postfix -- opendkim -e
```

8.3 Key Lessons Learned

1. **ConfigMap subPath mounts are cached** - Never use subPath for frequently updated configs
2. **Bake configs into Docker images** - More reliable for static configurations
3. **hostNetwork required for SMTP** - Port 25 needs node-level binding
4. **Delivery scripts must exist** - Postfix silently fails if mailbox_command path doesn't exist
5. **Queue delay matters** - Default 300s too slow for responsive email

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9. Appendices

Appendix A: File Structure

```
k8s/services/mail/
├── Dockerfile                  # RainLoop webmail image
├── nginx.conf                  # Nginx config for /mail/ path
├── php-fpm.conf                # PHP-FPM pool config
├── supervisord.conf            # Process supervisor config
├── mail-deployment.yaml        # Postfix + Dovecot deployment
├── mail-services.yaml          # ClusterIP services
├── mail-storage.yaml           # PersistentVolume + PVC
├── configs/
│   ├── postfix.yaml            # Postfix ConfigMap (legacy)
│   └── dovecot.conf.yaml       # Dovecot ConfigMap
├── domains/
│   └── shopfast.live.ini       # RainLoop domain config
├── dovecot/
│   └── Dockerfile              # Dovecot image
├── postfix/
│   ├── Dockerfile              # Postfix + OpenDKIM image
│   ├── main.cf                 # Baked-in Postfix config
│   ├── master.cf               # Baked-in master config
│   ├── opendkim.conf           # DKIM configuration
│   └── shopfast2026.private    # DKIM private key
└── rainloop-config/
    └── application.ini         # RainLoop application config
```

Appendix B: Environment Variables

Variable	Value	Used By
RAINLOOP_VERSION	1.17.0	Mail UI image
MYHOSTNAME	mail.shopfast.live	Postfix
MYDOMAIN	shopfast.live	Postfix

Appendix C: Resource Requirements

Component	CPU Request	CPU Limit	Memory Request	Memory Limit
Postfix	100m	250m	128Mi	256Mi
Dovecot	100m	250m	128Mi	256Mi
RainLoop (mail)	50m	250m	128Mi	256Mi

Appendix D: Version History

Version	Date	Changes
v1	Initial	Basic Postfix + Dovecot setup
v5	Added	DKIM signing via OpenDKIM
v10	Fixed	Domain config for RainLoop
v13	Fixed	RainLoop basePath and SSL settings
v20	Fixed	Added mbox-deliver script
v22	Fixed	Removed ConfigMap dependency
v25	Current	Baked config into image, 60s queue delay

Conclusion

The mail server implementation provides a complete email solution for the ShopFast platform. By baking configurations directly into Docker images and using a simple mbox-deliver script, the system achieves reliable email delivery with automatic queue processing every 60 seconds.

The architecture separates concerns: - **Postfix** handles SMTP reception and delivery - **Dovecot** handles authentication and IMAP access - **RainLoop** provides a modern webmail interface - **OpenDKIM** ensures email authentication

All components run as containers in Kubernetes with persistent storage, proper network configuration, and TLS security.